Module - 1

Introduction to Solar Energy Technology Systems

Solar energy is clean and renewable and there are many ways to use it. Solar energy can be:

- Converted to electricity
- Used for lighting
- Used for heating
- Used for cooling

Solar energy technologies include (Dutzik et al. 2010)

Photovoltaics (PV)

PV systems convert solar radiation into electricity. PV systems can take the form of panels or building materials. PV can be used in cold climates, have no “essential” moving parts, uses very little water, and are well suited for urban areas.

The most common forms of PV are crystalline silicone and “thin films”.

Crystalline silicone PV panels tend to be more expensive than “thin films” but are more efficient at converting electricity. These panels can be mounted on a roof, on top of a pole, or on a piece of machinery. An example of these panels can be seen below.

Thin films cost less and require less silicon. They can also be integrated into existing building features quite unobtrusively.

PV systems are easy to transport and install and can be used to generate electricity even in areas where there is no electrical grid.

PV systems are scalable and can be made to be as big as a power plant operation or as small as a hand-held calculator.

Furthermore, PV systems are excellent for urban areas. They produce no pollution and can be placed just about anywhere without interfering with human activity and do not require the acquisition of sites. In addition, because PV systems only require water for routine cleaning, they are suitable for areas with low water availability.

If every suitable rooftop in America were covered with PV panels, up to a quarter of America’s electricity could be produced by PV systems (Dutzik et al. 2010).

Concentrating Solar Power

Concentrating Solar Power (CSP) can provide a steady stream of power by storing the sun’s heat and using that heat to power a generator (Dutzik et al. 2010).

These systems use mirrors to focus light on a receiver to heat a fluid. This heated fluid is used to boil water. Once water is boiling, steam can be used to run turbines and generators.

Solar Water Heating

Solar water heating functions via sunlight absorbing pipes, which heat water as it travels. These systems can absorb as much as 87% of the energy that is radiated from a particular section of solar

collectors (Dutzik et al. 2010).

These systems can be used to heat swimming pools, water used in the home, or even the water for industrial use. These systems can often provide up to 50-80% of a building's water heating needs (Dutzik et al. 2010).

Active Solar Heating and Cooling

When using active solar heating/cooling systems, collectors are typically either glazed plates or evacuated tubes (Dutzik et al. 2010).

While most people understand how solar heating works, fewer understand how solar energy can be used to **cool** a building. The process can be broken down like this (Dutzik et al. 2010).:

1. Air is cycled through desiccant material (drying material, materials that work as a drying agent) that pulls humidity out of the air. One common example of a desiccant material is a silica gel packet, such is often found in the packaging of many products.
2. The desiccant material is then baked dry by solar heat.
3. The dry air can then be used to evaporate water from other sources.

This process is similar to the evaporation of sweat, a natural process that cools the human body. Active solar cooling systems are most effective on sunny days.

Passive Solar Lighting and Heating

In the past, most builders designed homes that took advantage of the sun to provide the dwelling with heat and light. However, during the twentieth century, builders relied upon the sun less and instead focused on cheap energy to provide heat and light to homes (Dutzik et al. 2010).

Using passive solar lighting is called **"daylighting"** and can reduce energy costs by as much as one third (Dutzik et al. 2010). This is a process that involves bringing a comfortable amount of light into a space without causing excessive glare or heat. Designers can use "windows, skylights, reflective surfaces, or exterior features that direct or diffuse light" (Dutzik et al. 2010, p. 14).

Passive heating, on the other hand, aims to design a building so that solar energy can be used when needed and can be blocked when it is not.

This involves orienting homes that admit winter sun but shade summer sun, including skylights, and encouraging good air circulation.

Furthermore, homes might be designed with sun spaces which provide a "pocket of warm air" outside the building or might include dark colored walls behind glass to conduct heat (Dutzik et al. 2010, p. 15).

What Solar Power Looks Like

Solar Homes

Homes can be built to efficiently harness the power of the sun, from passive design to PV panels. With solar energy, there is the potential of "zero net energy homes". A zero-net energy "produces as much energy as [it] consumes" (Dutzik et al. 2010 p. 2).

Current homes can be modified to use solar power as well, with an estimate of 35-40% of homes having the potential for PV panels and 50% of homes being suitable for solar heat collectors (Dutzik et al. 2010).

Like residential homes, commercial buildings also have the opportunity to make use of solar power. It is estimated that about 60-65 percent of commercial roof space is suitable for photovoltaics (Dutzik et al. 2010).

Some businesses have already taken advantage of the sun's natural resource. One example is Walmart's use of passive design, incorporating skylights in their stores to reduce the need for electric lighting.

Other businesses, such as the World's Largest Laundromat in Berwyn, Illinois, install solar water heating systems. This laundromat has installed a system with the capacity to produce 2,400 gallons of hot water a day.

Solar Factories

The creation of process heat requires more energy than any other activity in the manufacturing sector (Dutzik et al. 2010). **Process heating** refers to using heat to complete a process, such as cooking food or melting plastic.

Some of these processes require heat that is too high to be "economically" achieved by solar heat, but any process that occurs at less than 250 degrees Celsius would be well-suited to solar heat (Dutzik et al. 2010, p. 22). Worldwide, there are approximately 90 solar process heat systems in operation.

The largest of these systems is the Frito-Lay plant in California (Dutzik et al. 2010). This factory uses 5 acres of solar concentrators to provide heat to cook their food products, replacing the same amount of natural gas as the amount that is used in 340 average American homes.

Solar Farming

Following Franklin Delano Roosevelt's New Deal in the 1930s, the U.S. brought electricity to remote areas in rural America. This resulted in a shift from relying on the sun for certain agricultural processes to relying on electricity; one such example is crop drying.

While agricultural uses only result in 1% of America's energy consumption, the cost of fossil fuels can quite expensive for individual farmers. In 2003, farmers spent about 14% of their production expenses on energy (Dutzik et al. 2010).

Solar panels on farms could be used to: generate the electricity needed to operate a farm, keep crops cool, pump water, provide irrigation, heat greenhouses, ventilate barns, or even dry crops.

Solar in Transportation

There is a high dependence in American on petroleum, which has severe environmental, economic, and national security implications (Dutzik et al. 2010). Shifting American's focus off of petroleum and onto electricity would allow for more opportunities to use renewable energy, such as solar (Dutzik et al. 2010).

Furthermore, it is believed that the land used by America's transportation system could be used to harness the sun's energy. As of 2016, the Missouri Department of Transportation announced plans to begin development of solar roadways along Route 66.

Although electric vehicles may currently still be powered by the conventional grid, most cars simply sit still in a parking lot all day, making them an excellent candidate for solar energy in the form of solar charging stations.

Solar Communities

Solar power can be used anyplace where there is sunshine, so this included offices, schools, churches, and so on. Solar could be used to power district heating systems, district energy systems, and so on. Good candidates for solar energy include schools, churches, libraries, and museums, all of which frequently have large, flat roofs.

Many of these buildings use most of their energy for lighting or heating, both of which are easily provided by solar. Furthermore, most of these types of buildings will remain occupied by the same owners or activities for decades, making solar a smart investment down the line.

Solar Power Worldwide

Solar power and **solar energy** are two different things: solar power refers specifically to the conversion of the sun's rays into electricity, while solar energy refers more broadly to the sun's rays and heat, including passive solar design or solar water heating.

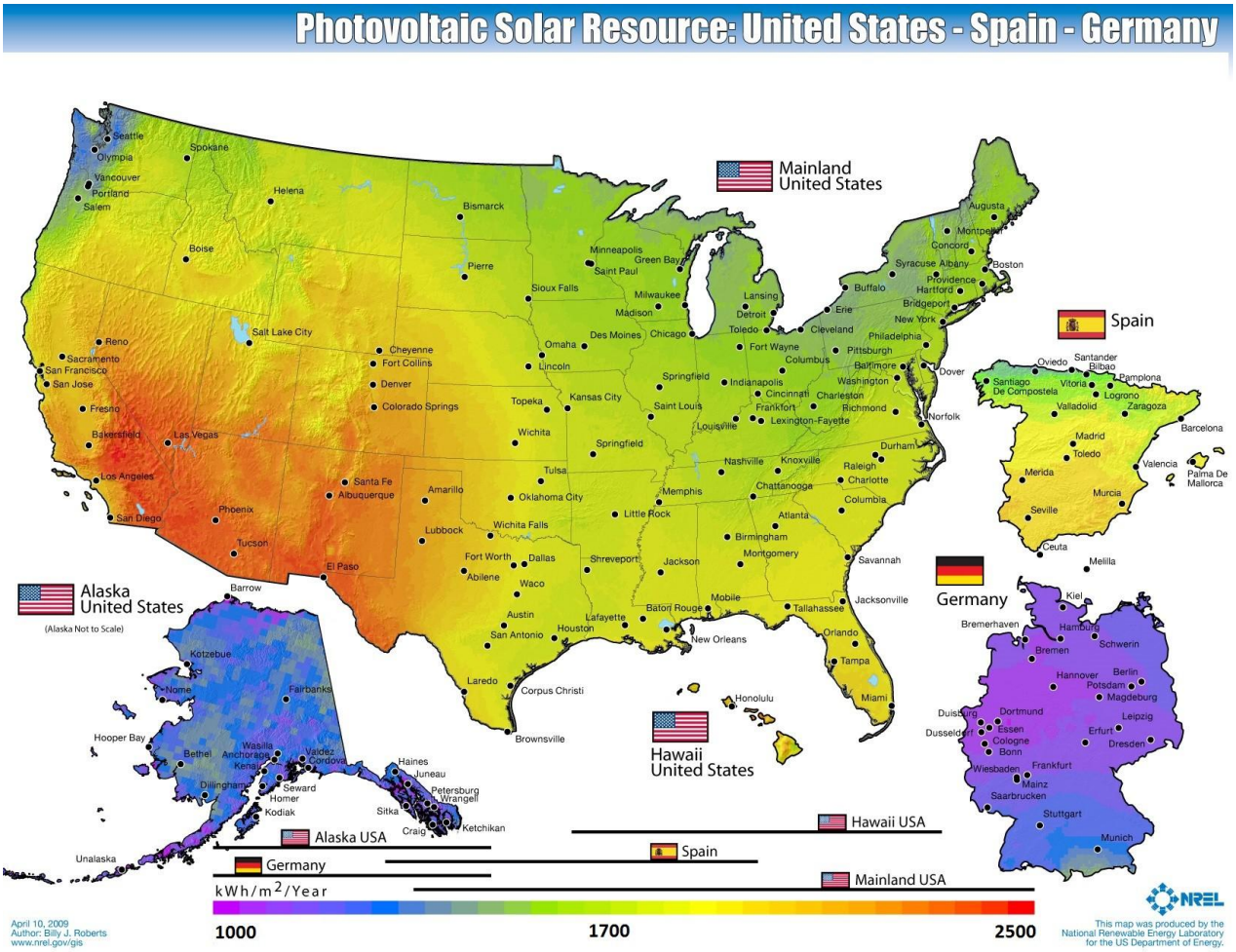
Solar power has been around for more than a century and was developed as early as the 1860s following the Industrial Revolution when people expected that coal was becoming scarcer. This led to the installation of the first PV system in New York in 1884. However, the oil boom in North America soon followed in the early 1900s.

It was not until the 1973 oil crisis that interest in renewable energy was revitalized.

Currently, the world leader in solar technology is the nation of **Germany** which, as one can imagine, is not the sunniest place in the world (Flannery and Sahajwalla 2013). Germany is a shining example of solar power efficiency even in places one does not typically associate with the sun.

As of 2008, Germany has an installed solar capacity of 32,200 MW (Flannery and Sahajwalla 2013). Solar capacity refers to the maximum power that can be instantly produced by a power plant.

With the map below, you can gain an understanding of the solar resources available in different parts of the world. Looking at the color spectrum key, you will see that nations that fall on the left side of the spectrum (purple and blue colors) have less solar resources while nations that fall to the right side of the spectrum (orange and red colors) have more solar resources.

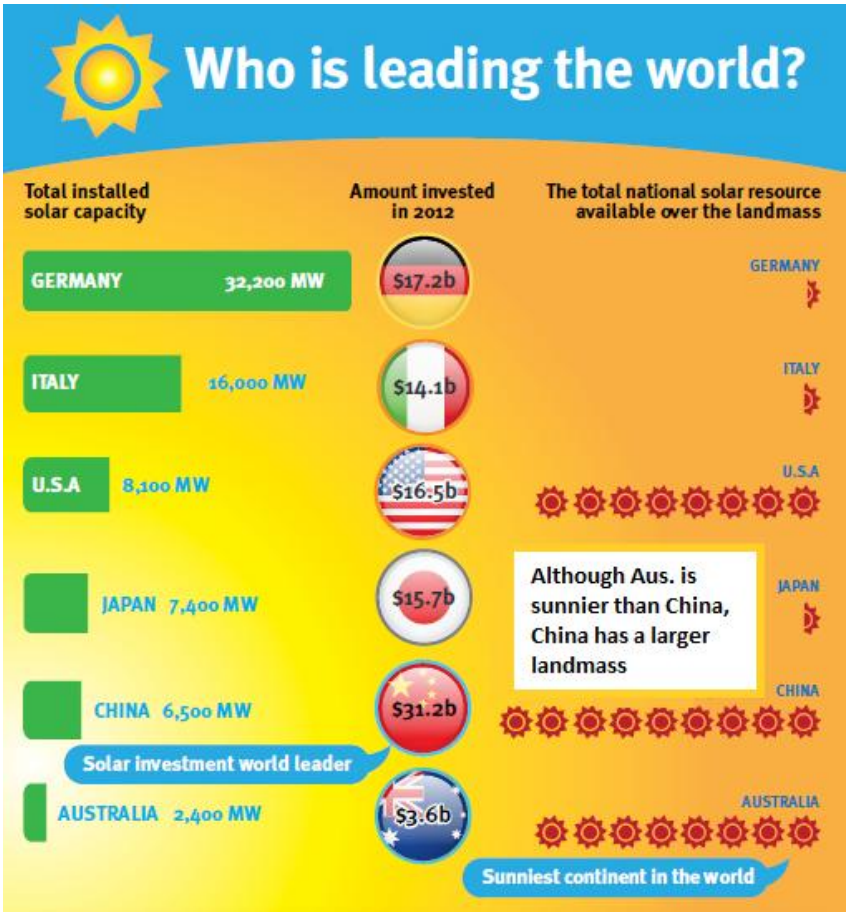


While Germany leads in installed capacity, the United States and China follow closely behind in terms of amounts of money invested in solar energy.

China has major concerns about pollution hopes to use renewable energy to reduce air pollution and greenhouse gas emissions (Flannery and Sahajwalla 2013).

The U.S., on the other hand, hopes to double the generation of electricity by 2020 (Flannery and Sahajwalla 2013).

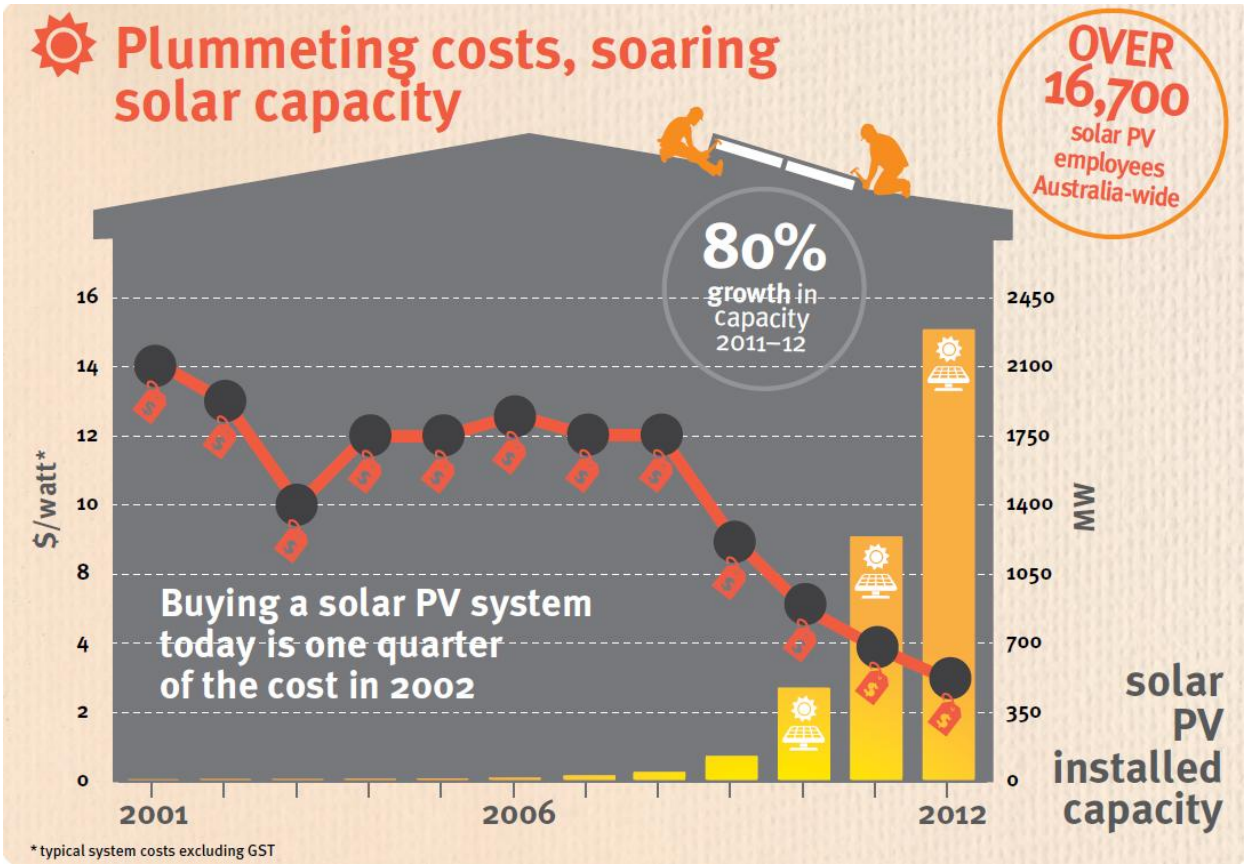
Australia, the world's sunniest continent, ranked 9th out of the G20 nations. The G20 is a forum composed of 20 of the world's major economies (Flannery and Sahajwalla 2013).



Solar energy has great prospects for the future. Coming up to 2013, the cost of buying a solar PV system was cut to a quarter of the cost it had been a decade before.

With the rate of cost declination of solar energy, it can be hoped that worldwide renewable energy capacity will continue to increase.

The graphic below shows the correlation between grid connected installed capacity and Australian dollars/watt over time.



The main points from this module are as follows:

Solar energy can be:

- Converted to electricity
- Used for lighting
- Used for heating
- Used for cooling

Solar energy technologies include:

- **Photovoltaics (PV)** – PV systems convert solar radiation into electricity.
- **Solar Water Heaters** - these rooftop collectors can capture solar energy and produce hot water.
- **Concentrating Solar Power (CSP)** – CSP plants function via mirrors that focus the suns energy to harness heat that can either be used directly or to generate electricity.
- **Active Solar Space Heating and Cooling** – Collectors like those used for heating water can also be used to heat air, replacing furnaces and boilers.
- **Passive Solar Design** – This refers to the design of homes and buildings that take advantage of natural sunlight to provide heat, light, and so on.

- **Homes** can be built to efficiently harness the power of the sun, from passive design to PV panels. With solar energy, there is the potential of "zero net energy homes". A zero-net energy "produces as much energy as [it] consumes" (Dutzik et al. 2010 p. 2).

- Like residential homes, **commercial buildings** also have the opportunity to make use of solar power. It is estimated that about 60-65 percent of commercial roof space is suitable for photovoltaics.
- Process heating refers to using heat to complete a process, such as cooking cement in a kiln, or melting iron in a mill (Dutzik et al. 2010). Some of these **factory** processes require heat that is too high to be "economically" achieved by solar heat, but any process that occurs at less than 250 degrees Celsius would be well-suited to solar heat (Dutzik et al. 2010, p. 22).
- Solar panels on **farms** could be used to: generate the electricity needed to operate a farm, keep crops cool, pump water, provide irrigation, heat greenhouses, ventilate barns, or even dry crops.
- Although electric **vehicles** may currently still be powered by the conventional grid, most cars simply sit still in a parking lot all day, making them an excellent candidate for solar energy in the form of solar charging stations.
- Good **community** candidates for solar energy include schools, churches, libraries, and museums, all of which frequently have large, flat roofs.
- Currently, the world leader in solar technology so the nation of Germany which, as one can imagine, is not the sunniest place in the world.
- While Germany leads in installed capacity, the United States and China follow closely behind in terms of amounts of money invested in solar energy.
- Australia, the world's sunniest continent, ranked 9th out of the G20 nations.

Module - 2

Why Solar and What is Preventing It?

84% of America's yearly energy use comes from fossil fuels (Dutzik et al. 2010).

Fossil fuels include fuels such as coal, oil, and natural gas. These sources are limited and expensive – American's spend approximately 9 cents of each dollar on the purchase of energy (Dutzik et al. 2010).

Aside from the cost of fossil fuels, they are extremely damaging to our environment and to our heath. Fossil fuels contribute to smog and soot, acid rain, and mercury poisoning of waterways (Dutzik et al. 2010). The drilling of fossil fuels often results in oil spills.

Many scientists fear that without a reduction of fossil fuel emissions, we may be reaching a "tipping point" in terms of global warming, after which mitigation will become much more difficult (Dutzik et al. 2010, p. 9).

Without positive change towards renewable energy, by the end of the century, the earth may see temperature increases of 11.5 degrees Fahrenheit and sea levels could rise by 6.5 feet.

But solar energy could change this. Even including the emissions created in the manufacturing of PV systems, life cycle emission reductions can be as high as 89 to 98 percent (Dutzik et al. 2010).

In 2004, one study estimated that if solar were utilized on all potential rooftops, approximately a quarter of American's electricity usage could be powered by solar energy (Dutzik et al. 2010).

That being said, there is “endless potential” to produce solar energy in particular locations of the States, particularly in places like the Southwest where the land is flat and the sun shines constantly.

Barriers to Solar Energy

Although the cost of solar power has dropped rapidly over the past few decades, there are still many barriers preventing the proliferation of solar energy. Barriers include:

Regulatory barriers: in some places, installing solar panels in one’s home or yard may be against the law.

Upfront cost: despite dropping installation costs and the lifelong investment, upfront costs are still high for families or small businesses.

Capital availability: It may be difficult for homeowners or business owners to obtain the credit required to finance the installation of solar energy systems.

Split incentives: when owners do not pay the utilities, they are likely to be less interested in installing an expensive solar energy system. Conversely, landlords are unlikely to consider solar technologies if they are unsure that they will receive the benefits of their purchase

Payback time: although a solar energy system may pay for itself over time, few are willing to invest in a technology that takes five or more years to pay back its costs (Dutzik et al. 2010).

Length of tenure: homeowners and business owners move frequently, thus making them wary of investing in a solar energy system that they may not be able to make full use of.

Failure to incorporate social benefits: despite contributing benefits to society as a whole, owners of solar panels rarely receive incentives.

Risk: there is less risk involved in drawing power from the current electrical grid in the event of a failure, as in the event of failure, costs will be covered by thousands or millions of consumers.

In the event of solar energy system failure, however, the cost will fall upon the system owner to repair or replace the failed system.

Utility barriers: utility companies may create problems for those who are trying to connect their solar energy systems to the electric grid, establishing limits on the systems eligible for compensation.

The Installation of PV Systems: An Overview

Before installing a PV system, many considerations must be made (Bilo et al. 2015):

Pre-Installation Consideration

- Check local building codes, subdivision covenants, zoning ordinances, and private easements
- Determine if a building permit is required
- Are there any historic district guidelines?
- Is there potential for future shading?

Who will install it?

Proper installation is a must – a qualified company or contractor is highly recommended.
Ensure that the company or builder that you have hired has experience in installing PV systems. To avoid being swindled, ask for the contact information of other customers so you can hear first-hand about the quality of services provided.
Do you need to hire a licensed contractor? Depending on where you live, you may need to hire a licensed professional if you want to connect your system to the utility's grid.
Are they insured for workmen's compensation in the case of an accident?
Check local listings in your country for qualified companies or contractors.
For example, the Clean Energy Project website can inform you about solar power installers in your area and decide which is right for you:
cleanenergyprojectbuilder.org



In regards to installation (handbook for solar photovoltaic (PV) systems 2009):
The ideal location varies greatly depending on what hemisphere you live in.
If you live in the Northern hemisphere, your solar panel should face south, and if you live in the Southern hemisphere, your solar panel should face true north.
If you live very close to the equator in a place like Singapore, you will be better off with a flat-mounted PV system.
Avoid shading, particularly shading from trees or neighboring buildings.
Do you want to incorporate solar panels aesthetically? Solar panels can be integrated into faces or incorporated into features such as canopies.

Following installation (Bilo et al. 2015):
Maintenance: While solar systems require less maintenance than systems with more moving parts, but they do still require routine maintenance.
Monitoring: If you want to ensure that your system is operating efficiently, monitoring devices can inform you about your system's power production.
Some monitoring systems include online access that can tell you about your system's performance, how much of your electricity was sent to the grid, your savings, and the amount of greenhouse gases that were not admitted (but would have been if using a non-renewable source).

How to Encourage Solar Energy

How can government, action groups, and communities encourage solar power? (Dutzik et al. 2010).

Government Backed Financial Incentives

- **Cash Incentives:** This refers to providing rebates to homeowners or businesses. These incentives can include rebates paid up front or over time, grants, or access to a renewable.
- **Tax Credits:** the current federal law allows for individuals to reclaim up to 30% of the cost of installing a “qualified solar PV or hot water systems” (Dutzik et al. 2010, p. 45).
- **Feed-In Tariffs:** These are rates that are paid to owners of PV systems that supply electricity to the grid.
- **Net-Metering:** Net metering would allow consumers to receive “fair compensation” for any solar electricity that they feed into the grid.

Putting Solar Energy on the Same Level as Traditional Forms of Energy

- **Putting a price on global warming pollution:** If the cost of energy reflects the cost of pollution caused by fossil fuels, this will encourage the development of clean energy as a long-term cheaper alternative.
- **Renewable energy standards (RES)** require that utilities in a certain state produce a particular amount of their energy from renewable sources.
- **Government purchasing requirements:** similar to RES, government purchasing requirements sees that the American federal government gets at least 7.5% of its electricity from renewable sources (this was the goal for 2013).

Overcoming the Cost

- **Municipal finances:** some municipalities now allowed to finance individual solar installations by assessing the owner’s tax bill. Home and business owners then pay back the cost over twenty years via their property taxes.
- **Community solar:** community solar panels allow individuals to purchase solar panels on utility land, thus investing in the success of solar energy.
- **Low interest loans and loan guarantees:** a low-interest loan allows homeowners to feel sooner the benefits of installing a solar energy system rather than having to pay a large amount of money up front.

Enacting Building Standards and Codes

- **Advanced building energy codes:** Energy codes are designed to improve the efficiency of buildings. In the search for energy efficiency and low lifelong costs, solar is favored with the goal of designing zero net energy buildings.

- **Government green building requirements:** These requirements ensure that all new public buildings are built to ensure maximum efficiency and the use of renewable energy.
- **Solar-ready homes policies:** These policies require all new homes to be designed to accommodate solar energy. If a home is built to use renewable energy, the cost of installation will not be so high.
- **Solar requirements:** To require something, such as the use of solar power, means that, inevitably, it will happen. This has been the case in Israel for decades, where solar water heaters are required. As a result, it is believed that 90% of Israeli homes use solar water heaters.

Guaranteed Access to Solar for All

- **Solar access laws:** Solar access laws could override homeowner's associations from placing restrictions on individual development of solar energy systems, as well as giving homeowners the "right" to sunlight if a tree or structure blocks the sun.
- **Permitting reforms:** To reduce or eliminate permitting fees will encourage the installation of solar energy systems as it will reduce the length of time it takes for individuals to gain approval.

Improving Education on Solar Power

- **Workforce training:** With the expanse of job training programs will ensure a "solar future" as more people become knowledgeable about solar energy and power.
- **Domestic manufacturing of solar energy equipment:** If solar energy demands continue to rise in America, the domestic production of solar energy equipment will expand the opportunities for work in the clean energy industry.
- **Building energy labeling:** just as there are government required EnergyGuide labels for appliances, new homes should be assessed for their energy efficiency so that individuals that build solar energy systems but then move will recapture the cost of their investment as the efficiency of the home would be reflected in the cost.
- **Public education programs:** Simply put, if the public is educated about solar powered technologies, they will have a better understanding of what types of solar power will be most efficient for their purposes.

The main points from this module are as follows:

- Without positive change towards renewable energy, by the end of the century the earth may see temperature increases of 11.5 degrees Fahrenheit and sea levels could rise by 6.5 feet.
- But solar energy could change this. Even including the emissions created in the manufacturing of PV systems, life cycle emission reductions can be as high as 89 to 98 percent.

Although the cost of solar power has dropped rapidly over the past few decades, there are still many barriers preventing the proliferation of solar energy. Barriers include:

- Length of tenure

- Split incentives
- Failure to incorporate social benefits
- Risk
- Utility barriers
- Regulatory barriers
- Upfront cost
- Capital availability
- Payback time

Considerations that must be made when installing a PV system:

- How experienced is your installer
- Where is the ideal location for your PV system
- Future maintenance and monitoring

Solar can be encouraged through:

- Government backed financial incentives
- Putting solar on the same level as traditional forms of energy
- Overcoming the cost
- Enacting building standards and codes
- Guaranteeing access to solar for all
- Improved education on solar